

Introduction to VPNs Theory

Task 1. In Cisco Packet Tracer, create two separate networks (representing two branch offices) using two routers and configure basic IP addressing so that each network can ping its own devices.

Two branch networks were built using the existing multi-area OSPF topology:

- Branch A: Router0, network 10.0.0.0/8, Switch0, PC0 (10.0.0.2), Area 1
- Branch B: Router1, network 30.0.0.0/8, Switch1, PC1 (30.0.0.2), Area 2
- Backbone link: Router0 Se0 (20.0.0.1) <-> Router1 Se0 (20.0.0.2), Area 0

Each branch was verified to reach its own devices (PC to default gateway, PC to switch-connected router) before the VPN was configured on top.

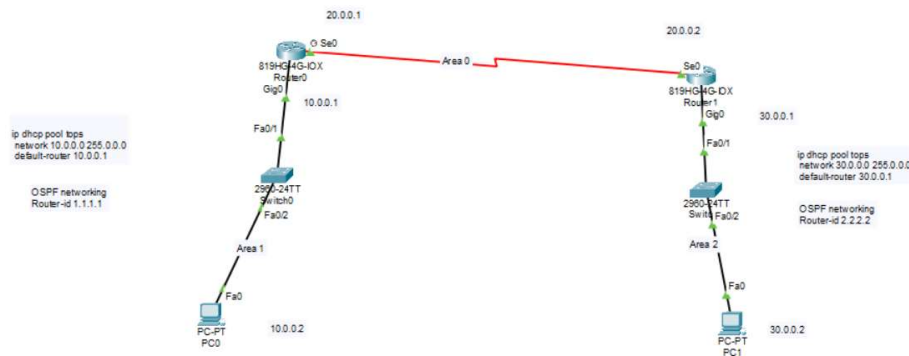


Figure 1: Two-branch topology used for the VPN lab (reused from the OSPF lab).

Task 2. Configure a basic site-to-site VPN between the two routers in Packet Tracer using GRE tunnel interfaces and assign tunnel IP addresses so that PCs from one branch can ping PCs in the other branch.

A GRE tunnel was built between Router0 and Router1, using their public-facing serial interfaces as the tunnel source/destination:

Router0 (Branch A):

```
enable
configure terminal
interface tunnel0
 ip address 172.16.1.1 255.255.255.252
 tunnel source Serial0
 tunnel destination 20.0.0.2
 no shutdown
exit
ip route 30.0.0.0 255.0.0.0 172.16.1.2
```

Router1 (Branch B):

```
enable
configure terminal
interface tunnel0
 ip address 172.16.1.2 255.255.255.252
 tunnel source Serial0
 tunnel destination 20.0.0.1
 no shutdown
exit
ip route 10.0.0.0 255.0.0.0 172.16.1.1
```

Verification: pinging the remote tunnel endpoint (172.16.1.2) succeeded with a TTL of 254, confirming the packet was freshly originated by the tunnel interface. A ping from PC0 (10.0.0.2) to PC1 (30.0.0.2) also succeeded with 0% loss, confirming the site-to-site VPN was passing traffic between branches.

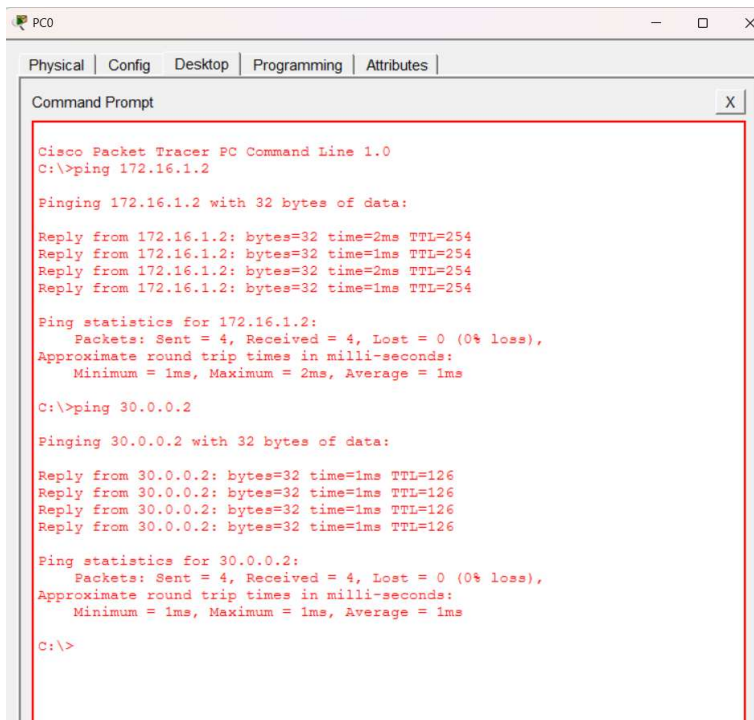
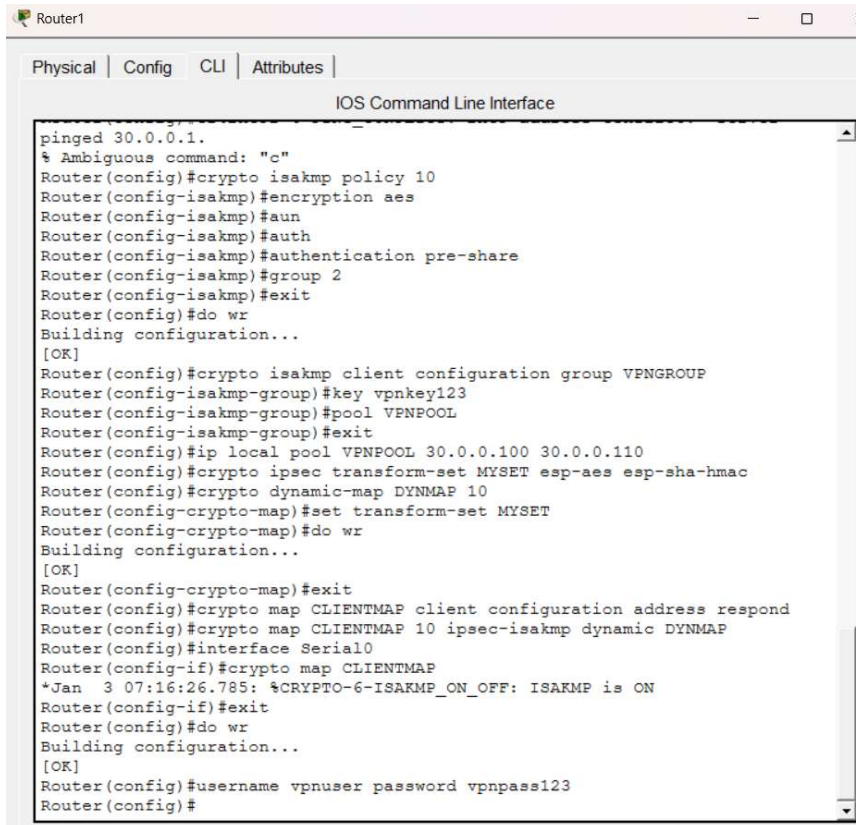


Figure 2: PC0 successfully pinging the tunnel endpoint (172.16.1.2) and PC1 (30.0.0.2) across the site-to-site VPN.

Task 3. Simulate a remote access VPN scenario in Packet Tracer by configuring a VPN client PC to connect securely to one of the branch routers using the built-in VPN client feature.

Router1 was configured as an Easy VPN server so that a remote PC (PC0) could connect into Branch B as if it were a remote employee:

```
enable
configure terminal
crypto isakmp policy 10
  encryption aes
  authentication pre-share
  group 2
exit
crypto isakmp client configuration group VPNGROUP
  key vpnkey123
  pool VPNPOOL
exit
ip local pool VPNPOOL 30.0.0.100 30.0.0.110
crypto ipsec transform-set MYSET esp-aes esp-sha-hmac
crypto dynamic-map DYNMAP 10
  set transform-set MYSET
exit
crypto map CLIENTMAP client configuration address respond
crypto map CLIENTMAP 10 ipsec-isakmp dynamic DYNMAP
interface Serial0
  crypto map CLIENTMAP
exit
username vpnuser password vpnpass123
aaa new-model
aaa authentication login VPNUSERS local
aaa authorization network VPNGROUP local
crypto map CLIENTMAP client authentication list VPNUSERS
crypto map CLIENTMAP isakmp authorization list VPNGROUP
end
write memory
```



```
Router1
Physical | Config | CLI | Attributes |
IOS Command Line Interface

pinged 30.0.0.1.
% Ambiguous command: "c"
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#aun
Router(config-isakmp)#auth
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 2
Router(config-isakmp)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#crypto isakmp client configuration group VPNGROUP
Router(config-isakmp-group)#key vpnkey123
Router(config-isakmp-group)#pool VPNPOOL
Router(config-isakmp-group)#exit
Router(config)#ip local pool VPNPOOL 30.0.0.100 30.0.0.110
Router(config)#crypto ipsec transform-set MYSET esp-aes esp-sha-hmac
Router(config)#crypto dynamic-map DYNMAP 10
Router(config-crypto-map)#set transform-set MYSET
Router(config-crypto-map)#do wr
Building configuration...
[OK]
Router(config-crypto-map)#exit
Router(config)#crypto map CLIENTMAP client configuration address respond
Router(config)#crypto map CLIENTMAP 10 ipsec-isakmp dynamic DYNMAP
Router(config)#interface Serial0
Router(config-if)#crypto map CLIENTMAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#username vpnuser password vpnpass123
Router(config)#
```

Figure 3: Easy VPN server configuration applied on Router1 (CLI).

On PC0, the built-in VPN client (Desktop > VPN) was configured with matching credentials:

- Group Name: VPNGROUP
- Group Key: vpnkey123
- Host IP (Server IP): 20.0.0.2
- Username: vpnuser
- Password: vpnpass123

Note: the first connection attempt returned "Connection timed out" because AAA (aaa new-model / authentication / authorization lists) had not yet been configured on the router, so the router had no way to validate the client's credentials during the handshake.

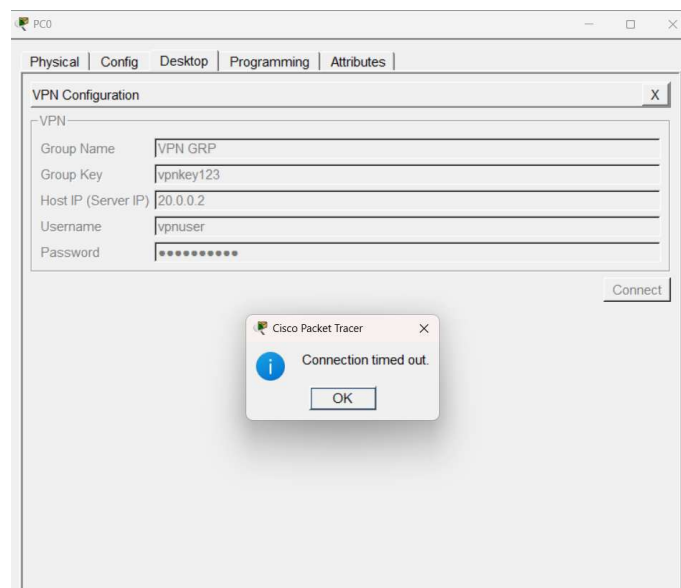


Figure 4: Initial connection attempt timing out before AAA was configured.

After adding the AAA configuration shown above, the VPN client connected successfully and was assigned an address from the configured pool:

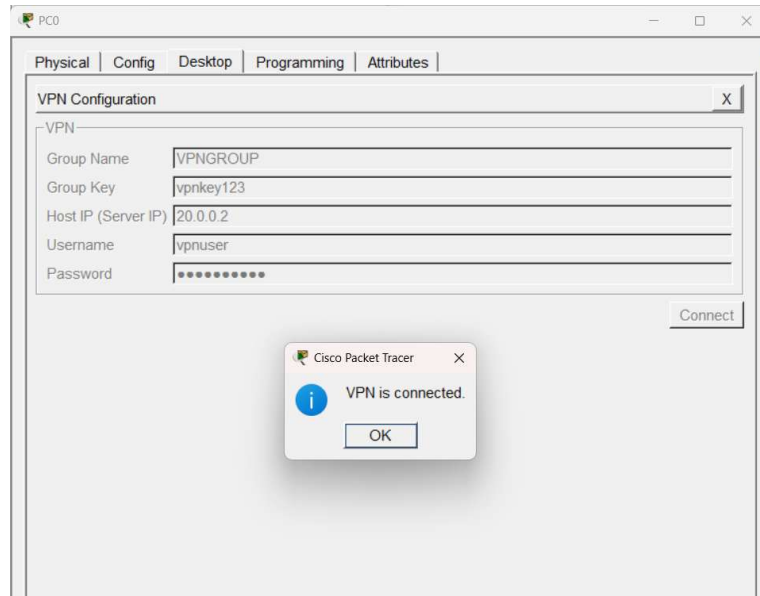


Figure 5: "VPN is connected" confirmation on PC0.

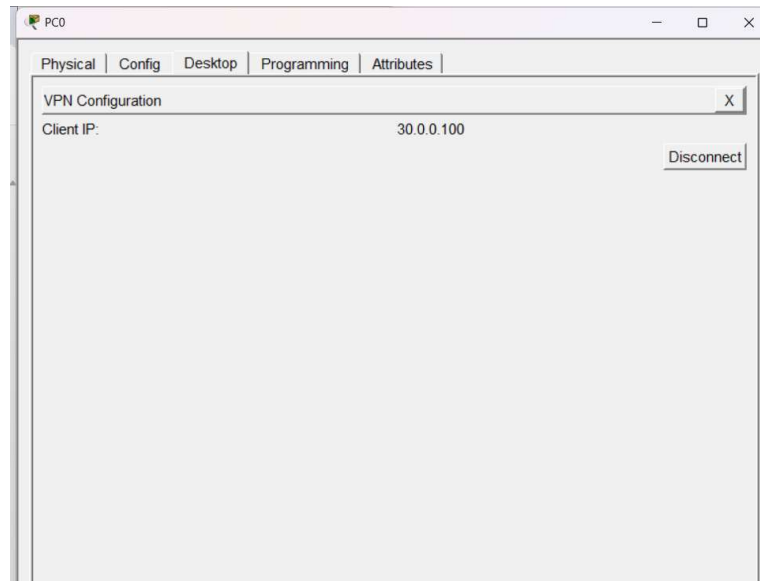


Figure 6: PC0 assigned Client IP 30.0.0.100 from the VPNPOOL range.

With the VPN active, PC0 could reach devices inside Branch B, confirming the remote access VPN was functionally live and not just cosmetically "connected":

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 30.0.0.1

Pinging 30.0.0.1 with 32 bytes of data:

Reply from 30.0.0.1: bytes=32 time=1ms TTL=255
Reply from 30.0.0.1: bytes=32 time=1ms TTL=255
Reply from 30.0.0.1: bytes=32 time=1ms TTL=255
Reply from 30.0.0.1: bytes=32 time=10ms TTL=255

Ping statistics for 30.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 3ms

C:\>ping 30.0.0.2

Pinging 30.0.0.2 with 32 bytes of data:

Reply from 30.0.0.2: bytes=32 time=11ms TTL=127
Reply from 30.0.0.2: bytes=32 time=14ms TTL=127
Reply from 30.0.0.2: bytes=32 time=12ms TTL=127
Reply from 30.0.0.2: bytes=32 time=16ms TTL=127

Ping statistics for 30.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 16ms, Average = 13ms

C:\>|

```

Figure 7: PC0 successfully pinging Router1 (30.0.0.1) and PC1 (30.0.0.2) over the remote access VPN.

4. List and explain the main differences between site-to-site VPN and remote access VPN based on your Packet Tracer configurations, including at least two advantages and disadvantages of each approach.

Aspect	Site-to-Site VPN	Remote Access VPN
Connects	Two networks / routers permanently	A single device to a network
Setup	Configured once on both routers (GRE tunnel)	Configured per connecting user/client
Advantage 1	Transparent to end users - no client software needed	Flexible - works from any single device, anywhere
Advantage 2	Efficient for constant, high-volume branch-to-branch traffic	Easy to add or revoke individual users without touching the branch config
Disadvantage 1	Inflexible - fixed to specific router locations	Depends on client-side VPN software/config on every device
Disadvantage 2	Both ends need static/known public IPs	Extra AAA overhead (authentication, authorization) per user

Task 5. Document and submit the step-by-step configuration commands you used to set up the site-to-site VPN in Packet Tracer, including interface, tunnel, and routing commands.

Router0 (Branch A) - GRE tunnel + routing:

```

enable
configure terminal
interface tunnel0
 ip address 172.16.1.1 255.255.255.252
 tunnel source Serial0
 tunnel destination 20.0.0.2
 no shutdown
exit
ip route 30.0.0.0 255.0.0.0 172.16.1.2

```

```
end  
write memory
```

Router1 (Branch B) - GRE tunnel + routing:

```
enable  
configure terminal  
interface tunnel0  
  ip address 172.16.1.2 255.255.255.252  
  tunnel source Serial0  
  tunnel destination 20.0.0.1  
  no shutdown  
exit  
ip route 10.0.0.0 255.0.0.0 172.16.1.1  
end  
write memory
```

Verification commands used:

- show ip interface brief (confirms Tunnel0 is up/up)
- show interface tunnel0 (confirms tunnel source/destination and GRE/IP transport)
- ping 172.16.1.2 (tests the tunnel interface itself)
- ping 30.0.0.2 from PC0 (tests end-to-end connectivity across the site-to-site VPN)